

ASSIGNMENT -2



K. R. MANGALAM UNIVERSITY

BTech CSE (AI & ML)

Section - B

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COURSE: Computer Networks

Course Code: (ENCS352)

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Assignment 2: Packet Flow Visualization Using Simulation Mode

Experiment Objectives

Students will use Packet Tracer Simulation Mode to observe how packets move between devices, how switches learn MAC addresses, and how ICMP echo request/reply travels through a small network.

Learning Outcomes

- Use Simulation Mode effectively
- Observe ARP + ICMP behavior
- Understand switch MAC learning behavior
- Interpret packet event list timeline

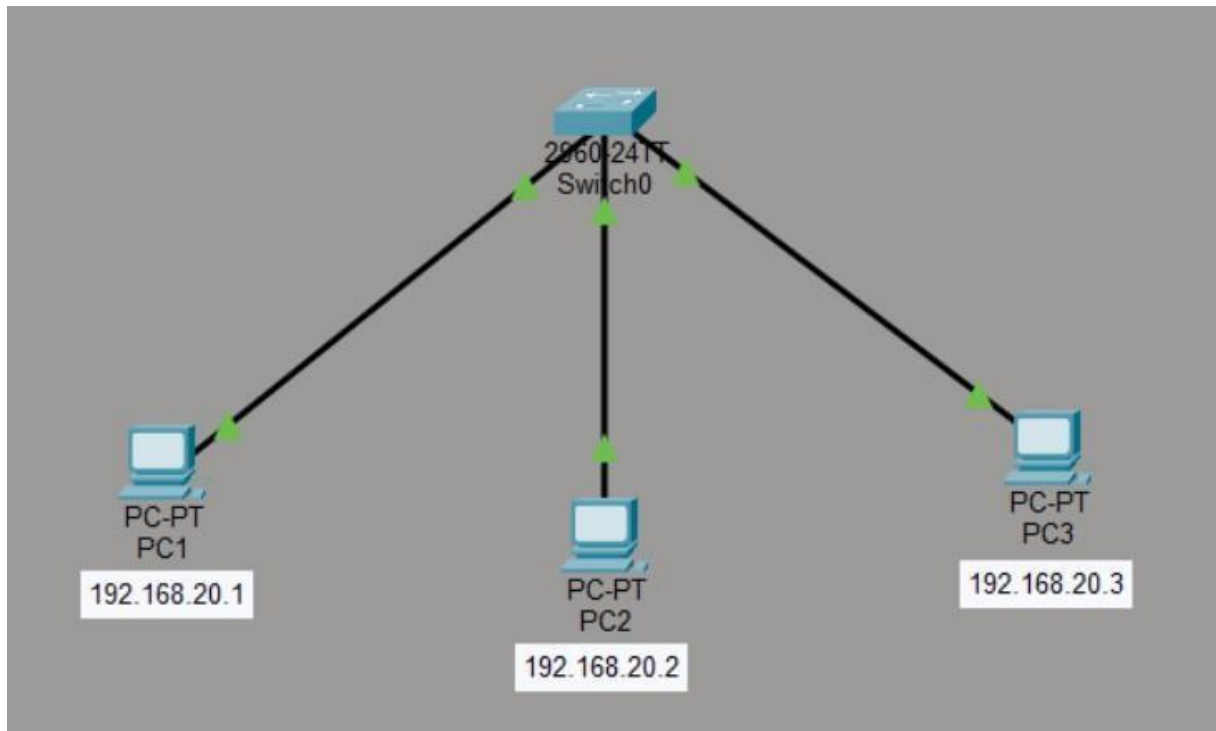
Concepts Used

- Simulation Mode
- ICMP ping
- ARP resolution
- Switch MAC address table learning

GITHUB LINK: <https://github.com/sj3101/CN-Assignment-2>

Task 1: Build a Simple LAN

- 1 switch + 3 PCs
- IP example: 192.168.20.1–3 /24



Observation

A simple LAN network was created using one switch and three PCs. Each PC was connected to the switch using Ethernet cables and assigned IP addresses in the same subnet (192.168.20.1–192.168.20.3 with subnet mask 255.255.255.0). The network topology was successfully established, allowing the devices to communicate with each other through the switch.

Task 2: First Ping Observation

- Enable Simulation Mode
- Ping PC1 → PC2
- Observe ARP request and ARP reply before ICMP

The screenshot displays a network simulation environment. On the left, a terminal window for PC1 shows the execution of the command `C:\>arp -a`, which returns "No ARP Entries Found". This is followed by a ping command `C:\>ping 192.168.20.2`. The output shows four successful replies from 192.168.20.2 with 32 bytes of data and a time of 4ms. Ping statistics for 192.168.20.2 indicate 4 packets sent, 4 received, and 0 lost, with round trip times ranging from 4ms to 8ms.

In the center, a network topology diagram shows a central switch labeled "Switch0" connected to three PCs: PC1 (192.168.20.1), PC2 (192.168.20.2), and PC3 (192.168.20.3). Below the diagram, an "ARP Table for PC1" is shown with the following entry:

IP Address	Hardware Address	Interface
192.168.20.2	0030 A399 589D	FastEthernet0

On the right, the "Simulation Panel" displays an "Event List" table with columns for Vis., Time(sec), Last Device, At Device, and Type. The events are as follows:

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC1	ICMP
	0.000	--	PC1	ARP
	0.001	PC1	Switch0	ARP
	0.002	Switch0	PC2	ARP
	0.002	Switch0	PC3	ARP
	0.003	PC2	Switch0	ARP
	0.004	Switch0	PC1	ARP
	0.004	--	PC1	ICMP
	0.005	PC1	Switch0	ICMP
	0.006	Switch0	PC2	ICMP
	0.007	PC2	Switch0	ICMP
	0.008	Switch0	PC1	ICMP
	0.185	--	Switch0	DTP
	0.186	Switch0	PC2	DTP
	1.009	--	PC1	ICMP
	1.010	PC1	Switch0	ICMP
	1.011	Switch0	PC2	ICMP

The simulation panel also includes a "Reset Simulation" button, a "Constant Delay" checkbox, and a "Captured to: 3.023 s" indicator.

Observation

During the first ping from PC1 to PC2 in Simulation Mode, an ARP request was broadcast by PC1 to discover the MAC address of PC2. PC2 responded with an ARP reply containing its MAC address. After the MAC address was learned and stored in the ARP table, ICMP echo request and echo reply packets were exchanged successfully between the two devices.

Task 3: Second Ping Observation

- Ping PC1 → PC2 again
- Observe fewer packets (ARP cached)

The screenshot displays a network simulation environment with three main components:

- PC1 Command Window:** Shows the execution of a ping command from PC1 to 192.168.20.2. The output indicates four successful replies with a 4ms round-trip time and 0% loss.
- Network Topology:** A central switch (Switch0) is connected to three PCs: PC1 (192.168.20.1), PC2 (192.168.20.2), and PC3 (192.168.20.3).
- ARP Table for PC1:** A table showing the IP address 192.168.20.2, Hardware Address 0030.A399.589D, and Interface FastEthernet0.
- Simulation Panel Event List:** A table listing network events. The relevant events for the second ping are:

Vis.	Time(sec)	Last Device	At Device	Type
	1.006	Switch0	PC2	ICMP
	1.007	PC2	Switch0	ICMP
	1.008	Switch0	PC1	ICMP
	1.951	--	Switch0	STP
	1.952	Switch0	PC3	STP
	1.952	Switch0	PC2	STP
	1.952	Switch0	PC1	STP
	2.012	--	PC1	ICMP
	2.013	PC1	Switch0	ICMP
	2.014	Switch0	PC2	ICMP
	2.015	PC2	Switch0	ICMP
	2.016	Switch0	PC1	ICMP
	3.017	--	PC1	ICMP
	3.018	PC1	Switch0	ICMP
	3.019	Switch0	PC2	ICMP
	3.020	PC2	Switch0	ICMP
	3.021	Switch0	PC1	ICMP

The event list shows that only ICMP echo request and reply packets were observed during the second ping, confirming that ARP was cached and no new ARP packets were generated.

Observation

During the second ping from PC1 to PC2, only ICMP echo request and echo reply packets were observed in the Simulation Event List. No ARP request or ARP reply was generated because the MAC address of PC2 was already stored in the ARP cache of PC1. This reduced the number of packets exchanged and made the communication faster.

Task 4: MAC Table Check

- View switch MAC table (if available) / observe switch learning through events

The screenshot shows a network switch CLI window titled 'Switch0' and a network diagram to its right.

CLI Window:

```
state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/3, changed state to up

Switch>enable
Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       000a.f345.8e62   DYNAMIC Fa0/1
1       0030.a399.589d   DYNAMIC Fa0/2
Switch#
```

Network Diagram:

A central switch labeled '26024T Switch0' is connected to three PCs:

- PC-PT PC1 (192.168.20.1) connected to Fa0/1
- PC-PT PC2 (192.168.20.2) connected to Fa0/2
- PC-PT PC3 (192.168.20.3) connected to Fa0/3

Observation

The MAC address table of the switch was checked using the `show mac address-table` command in the switch CLI. The table displays the MAC addresses of connected devices along with the ports through which they are connected. These entries are dynamically learned by the switch when it receives frames from the devices, allowing it to forward data efficiently to the correct port.

Short Notes: First Ping vs Second Ping

First Ping:

- Before sending data, the source device does not know the destination MAC address.
- An **ARP request** is broadcast to find the MAC address of the destination IP.
- The destination device sends an **ARP reply** with its MAC address.
- After that, **ICMP echo request and echo reply** packets are exchanged.
- Therefore, the first ping takes slightly more time and shows **ARP + ICMP packets**.

Second Ping:

- The destination MAC address is already stored in the **ARP cache**.
- No ARP request or ARP reply is needed.
- Only **ICMP echo request and ICMP echo reply** packets are sent.
- Communication is faster and involves **fewer packets**.